Python Programming
Unit 06 – Lecture 03 Notes
Pandas Basics (Series and DataFrame)

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Contents

1 Lecture Overview

Pandas is a powerful library for working with tabular data (CSV, Excel-like tables). It provides:

- **Series** (1D labeled data),
- **DataFrame** (2D table).

This lecture focuses on creating and manipulating these structures.

2 Setup (If Needed)

If Pandas is not installed:

```
pip install pandas
```

3 Core Concepts

3.1 Series

A Series is like a column with an index.

```
import pandas as pd

s = pd.Series([10, 20, 30], index=["a", "b", "c"])
print(s)
```

3.2 DataFrame

A DataFrame is a table of columns (each column can be considered a Series).

```
import pandas as pd

df = pd.DataFrame({
    "name": ["Asha", "Bilal", "Charu"],
    "marks": [88, 76, 91]
})
```

3.3 Inspecting Data

- `df.head()` shows first rows
- `df.tail()` shows last rows
- `df.info()` shows dtypes and missing data summary
- `df.describe()` gives statistics for numeric columns

3.4 Column Selection

```
marks = df["marks"]
```

3.5 Add/Delete Columns

```
df["cgpa"] = df["marks"] / 10
del df["cgpa"]
```

3.6 Boolean Filtering

```
passed = df[df["marks"] >= 50]
```

3.7 Iteration (Use Carefully)

Pandas is designed for vectorized operations. Iteration is slower but sometimes necessary. Common ways:

- `df.iterrows()`
- `df.itertuples()`

4 Demo Walkthrough

File: `demo/pandas_basics_demo.py`
Observe:

- creating a DataFrame,
- adding new derived columns,
- filtering using boolean expressions.

5 Interactive Checkpoints (with Solutions)

Checkpoint 1 Solution

Question: difference between Series and DataFrame?
Answer: Series is 1D labeled data; DataFrame is 2D tabular data with columns.

Checkpoint 2 Solution

Question: how to select `"marks"` column?
Answer: `df["marks"]`

6 Practice Exercises (with Solutions)

Exercise 1: Create a DataFrame

Task: Create a DataFrame with columns `name` and `age`.
Solution:

```
import pandas as pd
df = pd.DataFrame({"name": ["A", "B"], "age": [18, 19]})
print(df)
```

Exercise 2: Create `cgpa` Column

Task: Create `cgpa` from marks.

Solution:

```
df["cgpa"] = df["marks"] / 10
```

7 Exit Question (with Solution)

Question: one line to create `cgpa` from marks?

Answer: `df["cgpa"] = df["marks"] / 10`

Appendix: Slide Deck Content (Reference)

The material below is a reference copy of the slide deck content. Exercise solutions are explained in the main notes where applicable.

Title Slide

Repository: `https://github.com/tali7c/Python-Programming`

Quick Links

Core Concepts **Demo** **Interactive** **Summary**

Agenda

- Core Concepts
- Demo
- Interactive
- Summary

Learning Outcomes

- Explain what Pandas is used for
- Create and manipulate Series and DataFrames
- Use `head()`, `tail()`, column selection, and basic iteration
- Perform vectorized operations on columns

Why Pandas?

- Works with tabular data (like Excel/CSV)
- Powerful cleaning and transformation functions
- Easy grouping/aggregation and missing value handling

Series and DataFrame

- **Series**: 1D labeled array
- **DataFrame**: 2D table with labeled columns

Create a DataFrame

```python
import pandas as pd

df = pd.DataFrame({
    "name": ["Asha", "Bilal", "Charu"],
    "marks": [88, 76, 91]
})
```

Useful Methods

- `df.head(n)`, `df.tail(n)`
- `df.info()`, `df.describe()`
- Select a column: `df["marks"]`

Add/Delete Columns

```
df["cgpa"] = df["marks"] / 10
del df["cgpa"]
```

Vectorized Operations

- Column math applies element-wise (fast)

```
df["marks_plus_5"] = df["marks"] + 5
df["passed"] = df["marks"] >= 50
```

Demo: Series + DataFrame Operations

- File: `demo/pandas_basics_demo.py`
- Shows:
  - creating Series/DataFrame
  - head/tail
  - selecting and adding columns
  - basic boolean filtering

Checkpoint 1

Question: What is the difference between Series and DataFrame?

Checkpoint 2

Question: How do you select the `"marks"` column from a DataFrame?

Think-Pair-Share

Discuss:

- Why are vectorized operations preferred over loops in Pandas?

Key Takeaways

- Pandas simplifies tabular data handling
- Series is 1D; DataFrame is 2D
- Column operations are vectorized and efficient

Exit Question

Write one line to create a new column `"cgpa"` from `"marks"`.